

Technical Project Report - Android Module

VitalRoute

Subject: Universidade de Aveiro - Computação Móvel

Date: <date of completion>

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Project abstract: VitalRoute é uma aplicação de fitness e segurança pessoal que permite gravar atividades ao ar livre (corrida, caminhada e ciclismo). O elemento diferenciador desta aplicação é a parte da segurança: a aplicação monitoriza a atividade, deteta quedas ou imobilidade prolongada e envia alertas para contactos de emergência pré-definidos.

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1 Application concept

Overview

→ What is the concept of the application? E.g.: what is this app for? Who are the typical users? How would they benefit from using the app in their lives?

Essential journey map

→ discuss (at least) one user/customer journey map that is representative of your solution concept. A required touch point would be the app.

Note: this was outlined in classes and previously requested as homework....

2 Implemented solution

Technical design

Software architecture overview

App main modules, backend elements, external libraries or services dependencies, Connected devices (e.g.: read from sensors, NFC,...).

Interaction between the modules

Android app structure

Clarify the layers/components used to structure the project (e.g.: MVVM? Repository pattern? remote APIs mapping?...)

Detail, as applicable, how the Android Architecture Guidelines were implemented (ModelView, Flows, StateFlows, Coroutines, Room for local storage, etc.).

Don't need to go into detail, but explain as you would to another developer.

Data strategy

Explain the supporting data models/data structures, the data persistence & content update strategies (e.g.: Firebase collections, synchronization strategies for online/off line)

Advanced design strategies

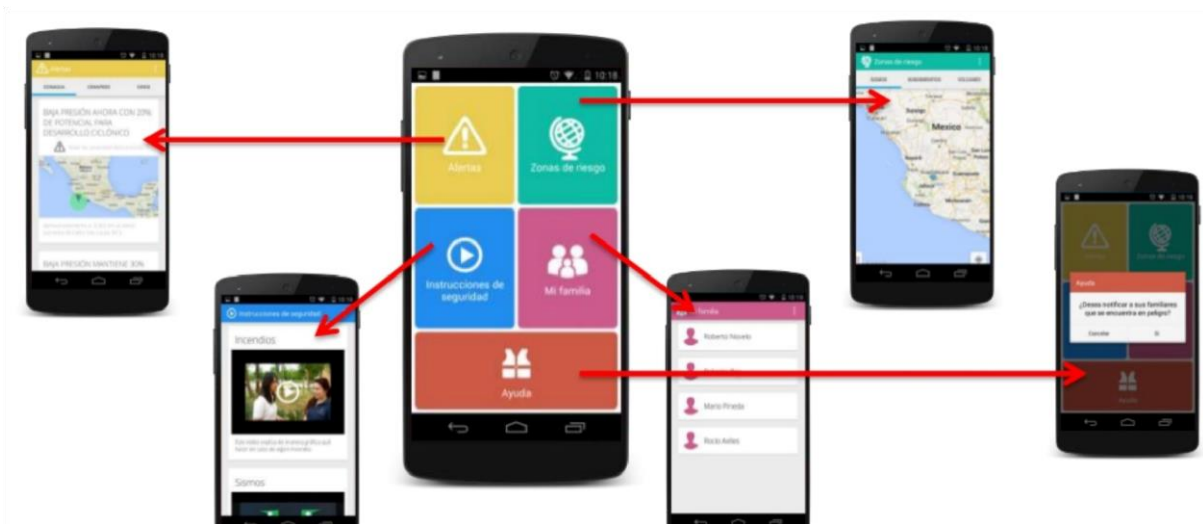
Project dependent. Highlights to the **hard parts**, if applicable.

e.g. which are the strategies for integration with Internet-based external services, reading from sensors, distributed workflows, deferred processing, push notifications mechanism,

distribution of updates to distributed devices, using AI for classification, find and connect to nearby devices, detect events in sensors, etc.

Implemented interactions

→ overview of the main interactions/usage flows; start with a “visual navigation map” with illustrative screenshots, and add concise supporting text to **explain the flow for the main “user stories”**. E.g.:



Project Limitations

<What is missing? Explain the **known limitations/unimplemented (but planned) features**>

New features & changes after the project presentation

<explain what new features/improvements were included in the project since the “official” project demonstration/oral presentation. Be specific.

Only applicable if you presented some time before the official submission deadline.>

3 Conclusions and supporting resources

Lessons learned

<describe the major “hard problems” found in the implementation process and how you solved those problems>

<if applicable: highlight aspects related to the Android development that were somewhat a surprise to the group, maybe because they were easy and powerful or, on the other hand, were harder than expected to tackle>

<if applicable: from your own experience, would you like to share some remark/suggestion with respect to the course “Computação Móvel” for future editions/future students?>

Work distribution within the team

<**required**: briefly describe the rationale for the work distribution in the group. Quantify the participation of each member adapting the next sentence >

Taking into consideration the overall development of the project, the contribution of each team member was distributed as follows: António Antunes did 60% of the work, and Belarmino Brito contributed with 40%.

Reference materials

<document the key components (e.g.: libraries, web API,...) or key references (e.g.: important blog post) used that were **really helpful** and certainly would help other students pursuing a similar work>

Project resources (direct links)

Resource:	Available from:
Code repository:	<URL to shared git repo; be sure that teacher can access!>
Ready-to-deploy APK:	<put URL for apk; may be inside the code repo; the debug apk, automatically generated, would be OK>
App Store page:	<put URL, only if applicable ; store can be Google Play, F-Droid,...>
Demo video:	<a link to a video demonstration of the app. Note mandatory, but consider make a video demo if your project requires specific/complex setup, not easily replicable.>